

AI-mazing responses: Elevating open-ended survey responses with generative AI follow-up questions

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Objective:

Explore whether generative AI can enhance the quality of open-ended survey responses by asking contextually relevant follow-up questions



Our motivation:



Why were you somewhat satisfied with your experience on National Airlines? Please share in detail your reason for selecting this rating.

The service was okay





:

fied with your experience
e in detail your reason for

The service was okay



Researcher reading
comments

With generative AI follow-ups

The service was okay



You mentioned that the service was 'okay'. Could you share what aspects of the service met your expectations and which aspects fell short?

Helping with overhead luggage compartment,
the beverages were too expensive



We'll share results from two studies

Study 1: Experimental study
(public LLM model)



Study 2: Observational study
(fine-tuned LLM model)



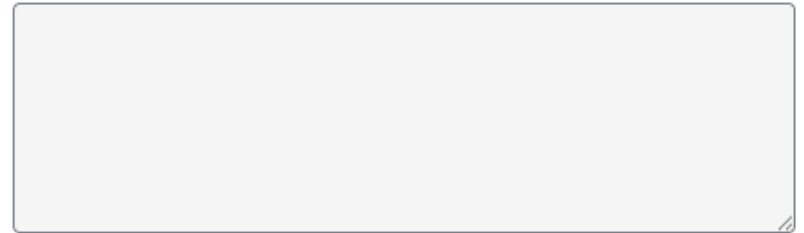
Image credit: Gemini

Study 1: Experimental Study

Study 1: Methods

- n=1,843; recruited from an online nonprobability panel in April and May 2024
- Quotas set for age, sex, race & education to reflect US population
- Topic: airline travel
- Model: GPT4-turbo
- Randomly assigned to 3 groups:
 - No follow-up
 - Up to 1 follow-up
 - Up to 2 follow-ups
- Follow-ups used on 2 OE questions →
- Analyzed lexical diversity, unique topics, and engagement

You said you were dissatisfied with your experience on National Airlines. Please share in detail your reason for selecting this rating.



How could National Airlines have improved your experience?



AI Prompt Engineering

NARROW, FOCUSED SCOPE

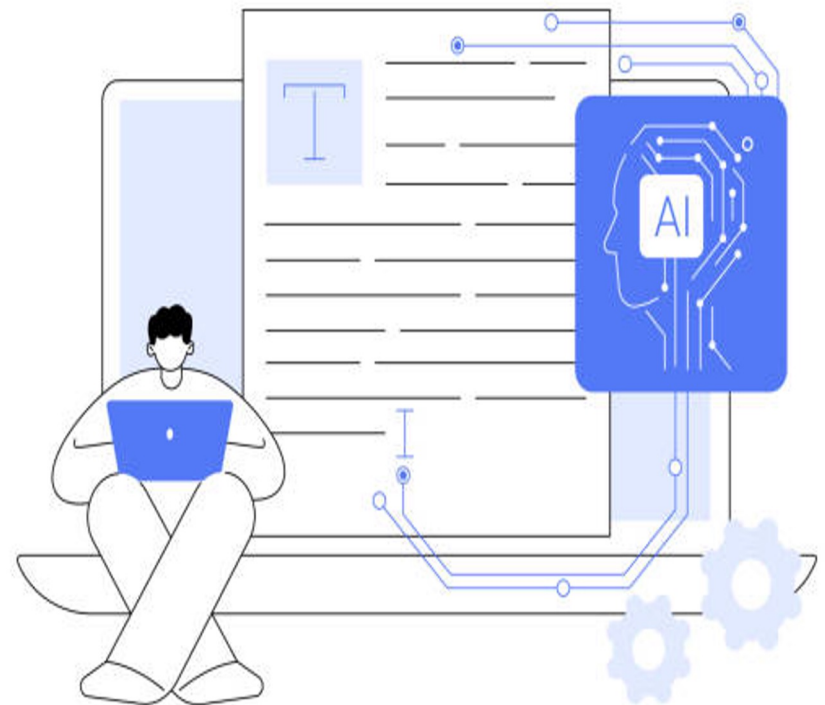
Designed to get more complete, detailed responses to original question. Does not ask net new questions.

FOLLOWS BEST PRACTICES

Incorporates best practices for questionnaire design and qualitative interviewing

ITERATIVE REFINEMENT

Prompt, annotate, evaluate, repeat. Learn what works well and what does not and iterate from there.



RESULTS: When generative AI follow-ups were used...

Respondents were more engaged

- Despite answering more questions, they found the survey **LESS burdensome to complete**
(9.7% AI vs 13.8% control, $p=0.0403$)
- **66%** of those who received a gen AI follow-ups said the survey was **MORE engaging** than other surveys
(31% said no difference)
- There was **no** statistically significant **impact on completion rates**
(97% AI vs 96% control)



"This AI chatbot is amazing, it's like I'm talking to you, but getting a sensible and considered response."

Responses were more insightful

- 87% more **topics** identified
(using TextIQ “Airlines” topic model)
- Detection of meaningful **sentiment, emotion, and effort** increased by at least **20 points** (from TextIQ)
- Percent of responses showing some **Lexical Diversity**¹ increased from 27% to 77%

¹McCarthy, P. M. & Jarvis, S. (2010). MTLT, vocd-D, and HD-D: A validation study of sophisticated approaches to lexical diversity assessment. *Behaviour Research Methods*, 42(2), 381–392.



What is lexical diversity?

- Lexical words are nouns, adjectives, verbs, and adverbs that convey the meaning of a text.
- Lexical words are divided by the total number of words – so longer responses CAN have lower lexical diversity than shorter responses



Study 2: Observational Study

Fine-Tuning

Adapts AI models for specific tasks like asking follow-up questions

1. Gather diverse training data
2. Annotate the data
3. Evaluate
4. Iterate
5. Get customer feedback
6. Repeat

Image credit: Gemini



Study 2: Methods

- Observational data selected from early adopters from November 24th to April 16, 2025
 - n =26,693 responses
 - n = 346 questions
 - n =128 surveys
- Data was anonymized and aggregated
- Mostly customer experience or market research across multiple industries (e.g., banking, retail, software)
- User settings:
 - # of follow-ups (up to 1 or 2) – we only analyze 1st follow up
 - Topics to avoid
- Use a fine-tuned model

**RESULTS: When generative AI follow-ups
(with fine-tuned model) were used...**

Participation remains high

- **91.6%** survey completion rate
(versus 84.3% for Qualtrics overall)
- **86.2%** of triggered follow-ups were answered
(versus 83.8% for the original open-ended question)
- Median time to load AI follow-up:
1.3 seconds

NOTE: Completion rates were updated post-conference due to an error in dates used for the initial calculation



Image credit: Gemini

Responses are twice as insightful



Measure	Original	Orig + AI	Improvement
Characters	87.0	184.9	2.1x
Total words	15.2	32.0	2.1x
Unique words	13.3	25.2	1.9x
Lexical diversity	24.4	48.0	2.0x
Lexical diversity > 0	39.8%	76.6%	1.9x
Avg TextIQ topics	3.0	6.5	2.2x
Industry topics (e.g., banking)	2.7	4.8	1.8x

Original response



Why did you cancel your membership with [organization]?

Very unresponsive



Lexical Diversity score = 0

Follow-up response



We understand that responsiveness is important. Could you provide an example of when you found the [ORGANIZATION] to be unresponsive?

We requested a different service but we did not receive a response. After a month, followed up by email and by phone. Still no response. Continued with email and phone calls for another 3 months until finally getting ahold of someone. Because of this, we are not going to continue our services



Lexical Diversity score = 98.3

Original + AI response

Very unresponsive



We requested a different service but we did not receive a response. After a month, followed up by email and by phone. Still no response. Continued with email and phone calls for another 3 months until finally getting ahold of someone. Because of this, we are not going to continue our services

Lexical Diversity score = 90.7

Limitations / Considerations

- Second study is observational, not experimental
- Analysis was not contextual (just looked at the response and not the question)
- AI follow-ups introduces the *potential* for bias. However...
 - Fine-tuning reduces this likelihood
 - Completion rates are high (not associated with higher drop-out rates)
 - Open-ended questions often toward the end of the survey

Conversational Feedback

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